

# Equity Autocalls and Vanna negative carries: Pricing and hedging with a simple add-on

Gregory Salon  
Quantik Trading Consulting  
gregory.salon@quantik.io

April 2019

## Abstract

### Autocall products: a toxic best-seller

Over the past twenty years, the autocall pay-off has been the most traded exotic equity product. Outstandingly popular, it is mainly sold to final customers in Europe and Asia through notes: its average yearly volumes reaches 100 billion euros. Nevertheless, it is responsible for major losses suffered by banks' Exotic trading desks. Roughly, when the spot remains in a [80%,105%] area around the recall barrier, daily carry loss is worth 1 to 3 basis points (depending on the product complexity, as several variations exist).

Indeed, it shows a strong model-dependency due to the cancellation feature: when the spot moves, books need dynamic rehedge via vanilla options, forward contracts, correlation products. As such, it requires the use of a pricing model which correctly combines market data dynamics (volatility, repo, equity correlations, quanto drifts...) and spot dynamic, in order to price properly the cost of daily rehedge.

In practice, building a pricing model complex enough to calibrate the relevant covariances while remaining numerically stable and computationally reasonable has proved to be a very serious challenge. Not mentioning ultimately the need for comprehensive interpretations of outputs. Escaping this issue, we exhibit here, a convenient way to price and hedge autocalls toxic behaviors through an additional and corrective pay-off.

There are approximations throughout the building of such an approach that have been tested numerically and justified qualitatively. Nonetheless, this is cheaper in terms of model complexity and development, and it provides a comprehensive and efficient pricing scheme combined with a hedging strategy which tackles the issue of negative carries generated by an autocall replication strategy.

## Summary

1. A standard autocall in a nutshell
2. Carry P&L of a pay-off
3. Autocall losses: a rough analysis of the famous spot/volatility negative carry
4. Price approximation of the cancellable put  $\Pi$
5. Approximation of Vanna derivatives
6. The model-dependent carry of an autocall
7. Autocalls in a local volatility model: a source of negative carries
8. A Trading recipe to price and hedge the model-dependent ExtraVanna P&L of autocalls
9. A standard 10 years autocall on SX5E: behaviors of  $ExtraVannaP\&L_{0 \rightarrow T}$
10. Conclusion

*Annex 1: Local volatility model, an embedded covariance between spot and volatility*

## 1. A standard autocall in a nutshell

We describe here a very common autocall; of course, there exists numerous variations inside the ‘autocall’ family.

The final client buys a note at inception  $t_0$  at 100 EUR

- If at an anniversary date  $t_i$ ,  $SX5E(t_i) > SX5E(t_0)$ ,  
The product is redeemed and the client receives  $100 + i \cdot \text{ContractualCoupon}$
- If at the maximum maturity  $T$  (example:  $T=10$  for 10 years), the product has never been recalled
  - If  $SX5E(T) > SX5E(t_0)$ , the client receives  $100 + T \cdot \text{ContractualCoupon}$
  - Else, the client receives  $100 - \text{European Down\&In Put (strike 100\% and activation barrier @ 50\%)}$

To simplify the exact nature of this product, the bank selling an autocall carries the following flows in its book:

- The bank is short a strip of bermudean digitals with an increasing height of coupon (and cancellable feature)
- The bank has a long position on a European Down&In Put (strike 100% and activation barrier @ 50%)  
which is cancellable every year  $t_i$ , if  $SX5E(t_i) > SX5E(t_0)$ .

For the clarity of the demonstration, one studies here a mono-underlying autocall in its domestic currency. As the volume traded is massive on SX5E, one contemplates a long term SX5E autocall, with a ‘European D&I put’ feature. Later in this article, we will generalize our conclusions to quanto multi-underlying autocalls.

## 2. Carry P&L of a pay-off

We consider here the replication P&L of an exotic pay-off  $P$  priced in a model (containing  $n$  market data  $[\alpha_i]_{1 \rightarrow n}$ ). Every 1<sup>st</sup> order derivative is considered as daily hedged.

The pay-off has a final term in  $T$ , the aggregated future Carry P&L is a random variable. Its value depends on the spread between the covariances that will occur in the future and the covariances embedded in the pricing model.

$$\text{CarryP\&L}_{0 \rightarrow T} = \frac{1}{2} \int_0^T \left[ \sum_{i,k} \alpha_i \alpha_k \frac{\partial^2 P}{\partial \alpha_i \partial \alpha_k} \Big|_{t, \alpha_i(t), \alpha_k(t)} [\text{InstaRealisedCovar}_t(d\alpha_i, d\alpha_k) - \text{InstaModelCovar}_t(d\alpha_i, d\alpha_k)|_t] \right] \quad (1.)$$

Two situations may occur:

- The pay-off  $P$  in the replicating portfolio is priced and traded in the market, which means that:
  - $P$  is model-independent; said differently, there is a static replication of  $P$ , with instruments available in the market
  - The valuation of  $P$  is consistent with calibrated market quotes and, by essence,  $E[\text{CarryP\&L}_{0 \rightarrow T} | \mathcal{F}_0] = 0$
- $P$  has to be dynamically hedged with other instruments, therefore, it is a model-dependent pay-off:
  - The model has to diffuse market data involved in non-zero 2<sup>nd</sup> derivative (or relevant method: e.g. quanto drift)
  - Model covariances linked to the above data need to be chosen according to a statistical study of the past realized covariances
  - The valuation of  $P$  depends on the describing span of the model, and relies on the relevant feeding of model covariances

### 3. Autocall losses: a rough analysis of the famous spot/volatility negative carry

For an autocall, the Put, with his European Down&In and yearly bermudean Up&Out features, explains the most part of the negative carries.

Indeed, the cancellation feature of the D&I Put implies that there is no static replication: every spot move requires a partial rehedge through an appropriate quantity of a vanilla like D&I Put. Of course, cancellable digital coupons imply also a dynamic rehedge: one could extend the demonstration to these flows as well.

Let's call  $\Pi$ , the cancellable D&I Put, which is bought and then replicated by the bank. It is the core component of the autocall, which illustrates the model dependency.

If the spot drops, the recall probability drops also and the Vega of  $\Pi$  increases. The trader has to sell vanilla options in order to flatten his Vega sensitivity (if the spot rises, one gets the reverse situation):  $\frac{\partial^2 \Pi}{\partial S_t \partial \sigma_{atm}} < 0$

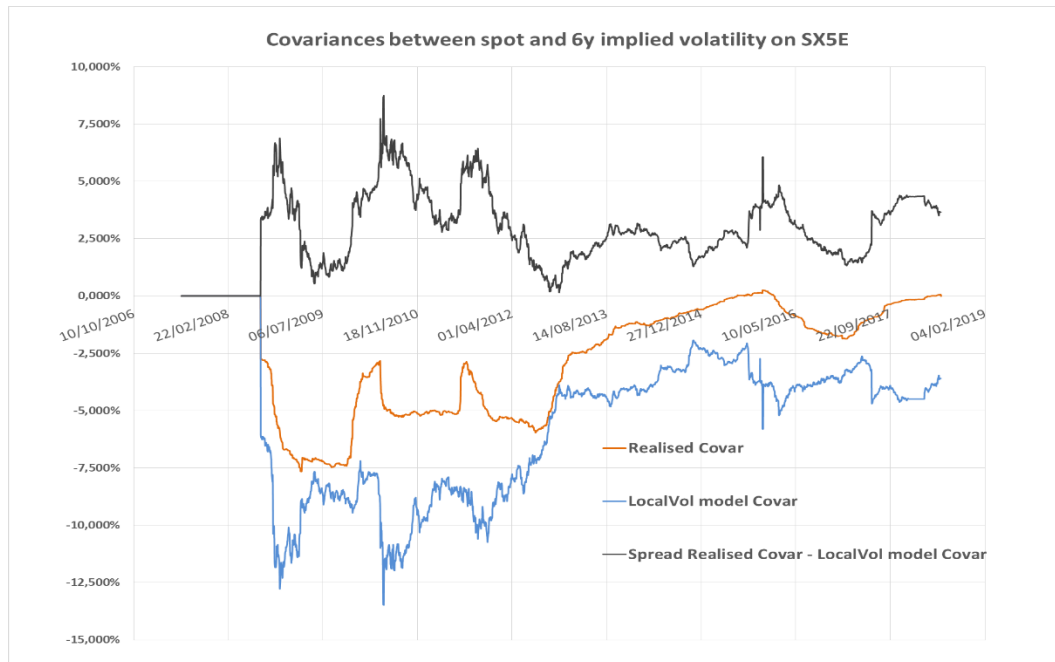
One understands easily the importance of the combined dynamic between volatility and spot. As soon as the  $ModelCovar_{0 \rightarrow T}(dS_t, d\sigma_{implied})$  is in line with the realized covariance, the Carry P&L spot/vol is flat.

The Vanna spot/vol P&L term between  $t$  and  $t+dt$  is:

$$CarryP\&L_{t \rightarrow t+dt, S_t/\sigma_{atm}} = 2 \cdot \frac{1}{2} \frac{\partial^2 \Pi}{\partial S_t \partial \sigma_{atm}} \Big|_{t, S_t, \sigma_{atm}} [InstaRealisedCov(dS_t, d\sigma_{atm}) - InstaModelCov(dS_t, d\sigma_{atm})|_t] \quad (2.)$$

Unfortunately, the local volatility model, so convenient by many aspects, provides through its intrinsic calibration a value for the model covariance spot/vol which is extremely different from realized covariance.

$ModelCovar(dS_t, d\sigma_{atm})$  is calculated using a formula demonstrated in Bergomi Lorenzo, *Stochastic Volatility Modeling*, Chapman and Hall, 2016, p. 51. It has been reproduced in Annex 1.



- $\frac{\partial^2 \Pi}{\partial S_t \partial \sigma_{atm}}$  is negative
- $[RealisedCovar(dS_t, d\sigma_{atm}) - LocalVolModelCovar(dS_t, d\sigma_{atm})|_t]$  is positive on historical records.

Then  $CarryP\&L_{t \rightarrow t+dt, S_t/\sigma_{atm}}$  in past realisations has always been negative.

**This emphasizes genuinely the constant negative carry on the Vanna spot/volatility term for autocalls books managed with a local volatility model.**

#### 4. Price approximation of the cancellable put $\Pi$

$\Pi$  could be rewritten as a regular Down&In Put of maturity T multiplied by a **survival event** in T:

$$\begin{aligned}\Pi(t, T) &= Put(Spot_T, K_{put}, Barrier_{down\&in}, Cancellation_{level}, T) \\ &= Put_{Down\&In}(S_T, K, Barrier, T) \cdot I[\forall t_i, S_{t_i} < Cancellation_{level}] \\ &= Put_{Down\&In}(S_T, K, Barrier, T) \cdot I_{survival, T}(\{S_{t_i}\})\end{aligned}\quad (3.)$$

One would like to approximate, for any future pair  $(t, S_t)$ , e.g. locally, the value of the cancellable put.

**One uses here the following approximation:**

$$E[\Pi|_{t, S_t}] = Price_{\Pi}(t, S_t, T) \approx E[Put_{Down\&In}(S_T, K, Barrier, T)|_{t, S_t}] \cdot E[I_{survival, T}(\{S_{t_i}\})|_{t, S_t}]$$

It means that we put aside the covariance between the terminal put (in T) and the survival event (mostly dependent on shorter terms). This pricing error between this proxy and the true valuation is weak, especially when T is above 3 years. Moreover, the neglected covariance is the spot/spot one, which is not responsible for P&L leakage.

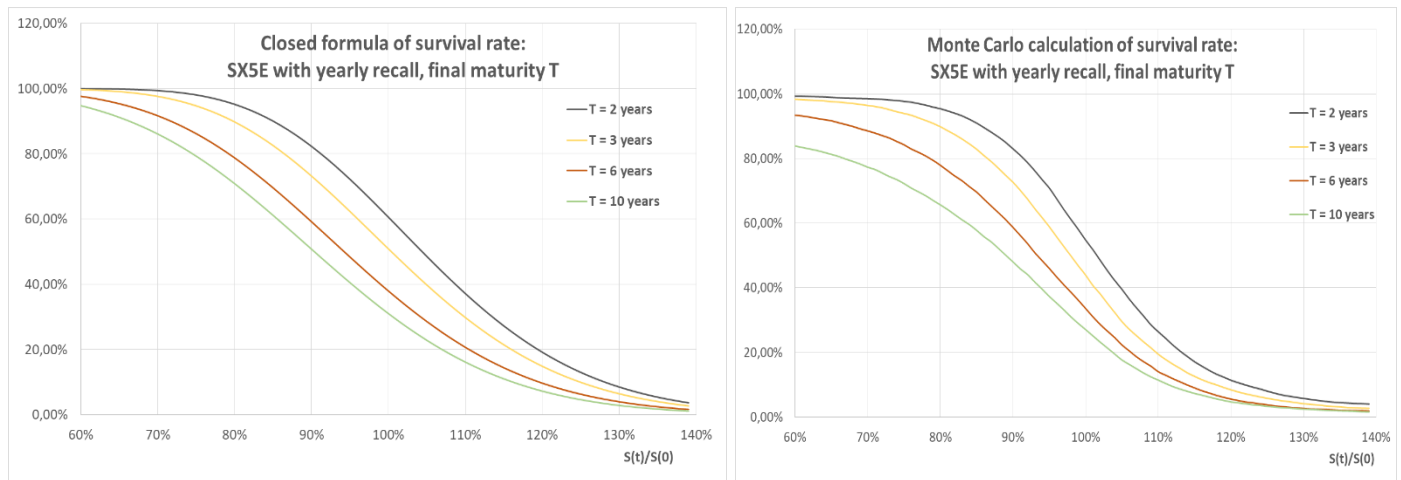
This proxy brings simplifications:

- There is a closed formula for a vanilla option such as  $E[Put_{Down\&In}(S_T, K, Barrier, T)|_{t, S_t}]$   
That we note **PutD&IPrice** $(t, S_t, T)$  and it is available for any pair  $(t, S_t)$ .
- There is a closed formula for  $E[I_{survival}(\{S_{t_i}\})|_{t, S_t}]$  (product of European digitals under conditional probability)  
That we note **Survival<sub>rate</sub>** $(t, S_t, T)$  and it is available for any pair  $(t, S_t)$ .

Therefore, one gets:

$$Price_{\Pi}(t, S_t, T) = PutD\&IPrice(t, S_t, T) Survival_{rate}(t, S_t, T) \quad (4.)$$

**Comparison** between the Monte Carlo valuation of the Survival rate and its closed formula:



## 5. Approximation of Vanna derivatives

As said in chapter 3., the dynamic reheding is a consequence of spot moves. Then significant derivatives  $\frac{\partial^2 \Pi}{\partial \alpha_i \partial \alpha_k} \Big|_{t, \alpha_i(t), \alpha_k(t)}$ , involved in the Carry P&L (formula (1.)), are those with a derivative regarding  $S_t$ . Indeed the survival event is strongly affected by the spot move and strengthens the 2<sup>nd</sup> derivative  $\frac{\partial^2 \Pi}{\partial S_t \partial \alpha_k} \Big|_{t, S_t, \alpha_k(t)}$

$\frac{\partial^2 \Pi}{\partial S_t \partial \alpha_k}$ , called **Vanna derivatives**, are responsible for the negative carries during the replication of autocalls.

From now on, the Carry P&L between  $t$  and  $t+dt$ , for any  $(t, S_t)$ , applied to the replication of a cancellable put  $\Pi$ , will be reduced to the Vanna P&L terms:

$$VannaP\&L_{t \rightarrow t+dt} = \sum_{\alpha_k} \frac{\partial^2 \Pi}{\partial S_t \partial \alpha_k} \Big|_{t, S_t, \alpha_k(t)} [RealisedCov(dS_t, d\alpha_k) - ModelCov(dS_t, d\alpha_k)|_t] \quad (5.)$$

In chapter 3., one focused on the Vanna spot/vol. Spot/dividend, spot/repo or spot/quanto drift dynamics are affected in the same way. These are also responsible for daily negative carries in autocalls books.

Regarding any  $\alpha_k$ , one can derivate the approximate closed formula of  $\Pi$ , (4.) :

$$\begin{aligned} \frac{\partial^2 Price_{\Pi}(t, S_t)}{\partial S_t \partial \alpha_k} &= \frac{\partial^2 [PutD\&IPrice(t, S_t). Survival_{rate}(t, S_t)]}{\partial S_t \partial \alpha_k} \\ &= \frac{\partial PutD\&IPrice}{\partial S_t} \frac{\partial Survival_{rate}}{\partial \alpha_k} + \frac{\partial Survival_{rate}}{\partial S_t} \frac{\partial PutD\&IPrice}{\partial \alpha_k} \\ &\quad + \frac{\partial^2 Survival_{rate}}{\partial S_t \partial \alpha_k} . PutD\&IPrice + \frac{\partial^2 PutD\&IPrice}{\partial S_t \partial \alpha_k} . Survival_{rate} \end{aligned}$$

**1<sup>st</sup> simplification:** negligible terms

$\frac{\partial PutPrice}{\partial S_t} \frac{\partial Survival_{rate}}{\partial \alpha_k}$  and  $\frac{\partial^2 Survival_{rate}}{\partial S_t \partial \alpha_k} . PutPrice$  are negligible as a numerical study shows (not displayed here).

**2<sup>nd</sup> simplification:** keeping only the exotic term

$\frac{\partial^2 PutPrice}{\partial S_t \partial \alpha_k} . Survival_{rate}$  should not be taken into account because it is already taken into account in vanilla P&L

- From  $t$  to  $t+dt$ , this greek is hedged through a short position of a vanilla D&I Put for a quantity equal to  $Survival_{rate}$ .
- $\frac{\partial^2 PutPrice}{\partial S_t \partial \alpha_k} . Survival_{rate} . [RealisedCov(dS_t, d\alpha_k) - ModelCov(dS_t, d\alpha_k)|_t]$  is already part of the daily P&L of a calibrated vanilla options and for which,  $E[CarryP\&L_{0 \rightarrow T} | \mathcal{F}_0] = 0$  (see chapter 2.)

Providing that we want to extracts the part of the carry P&L which is purely exotic, one gets rid of this term.

Thus, Vanna intensity of a D&I Put cancellable  $\Pi$  (for its model-dependent part) gets a good proxy with:

$$\frac{\partial^2 Price_{\Pi}(t, S_t, T)}{\partial S_t \partial \alpha_k} \approx \frac{\partial Survival_{rate}}{\partial S_t} \frac{\partial PutD\&IPrice}{\partial \alpha_k} \quad (6.)$$

## 6. The model-dependent carry of an autocall

The formula (6.) is tied to an **intuitive replication** (even if not purely exact) **for a long position of  $\Pi$** .

- In  $(t, S_t)$ , the trader holds a short position of vanilla D&I Put for a quantity equal to  $Survival_{rate}(t, S_t, T)$ .
- In  $(t + dt, S_{t+dt})$ , if  $Survival_{rate}(t + dt, S_{t+dt}, T)$  has changed, the trader has to buy or sell a certain amount of vanilla D&I Put, in order to rehedg his book. This variation of quantity is equal to  $\frac{\partial Survival_{rate}}{\partial S_t} \cdot dS_t$
- This rehedg implies a “cross gamma  $dS_t, d\alpha_k$ ” P&L:  $\sum \alpha_k \frac{\partial Survival_{rate}}{\partial S_t} \frac{\partial PutD\&I Price}{\partial \alpha_k} dS_t \cdot d\alpha_k$   
Compensated by a “theta P&L:  $-\sum \alpha_k \frac{\partial Survival_{rate}}{\partial S_t} \frac{\partial PutD\&I Price}{\partial \alpha_k} ModelCov(dS_t, d\alpha_k)$

Locally, for any  $(t, S_t)$ , the **daily ExtraVanna P&L** for its model-dependent part (‘Extra’ emphasizes this point), of the book replicating  $\Pi$  is:

$$ExtraVannaP\&L_{S_t, t \rightarrow t+dt} = I|_{no\ recall\ before\ t} \sum \alpha_k \frac{\partial Survival_{rate}(t, T)}{\partial S_t} \frac{\partial PutD\&I Price}{\partial \alpha_k} \cdot [RealisedCov(dS_t, d\alpha_k) - ModelCov(dS_t, d\alpha_k)|_t] \quad (7.)$$

*Nota bene:*

$Survival_{rate}(t, T)$  is a closed formula (see chapter 4.), calculating in  $t$  the survival probability in  $T$ , which assumes that the product has not been called before  $t$ .  $I|_{no\ recall\ before\ t}$  is therefore introduced to get the proper expression of  $ExtraVannaP\&L_{S_t, t \rightarrow t+dt}$ , random variable seen from  $t=0$ , conditionally to  $(t, S_t)$ .

One can write **the model-dependent  $ExtraVannaP\&L_{0 \rightarrow T}$**  (random variable) of a short position on autocall, which is the same as a long position on a Cancellable Put:

$$ExtraVannaP\&L_{0 \rightarrow T} = \int_0^T I|_{no\ recall\ before\ t} \sum \alpha_k \frac{\partial Survival_{rate}(t, T)}{\partial S_t} \frac{\partial PutD\&I Price}{\partial \alpha_k} \cdot [RealisedCov(dS_t, d\alpha_k) - ModelCov(dS_t, d\alpha_k)|_t]$$

Or in a more contracted way

$$ExtraVannaP\&L_{0 \rightarrow T} = \int_0^T ExtraVannaP\&L_{S_t, t \rightarrow t+dt} \quad (8.)$$

*Nota bene:*

This formula could be easily extended to multi-underlying products, especially to take into account the covariance between spot and equity correlations, in the same way the covariance between spot and volatility is treated here.

Providing that autocall model-dependency relies mostly on rehedging due to spot moves, one has neglected other terms which could bear Extra Carries. For instance,  $voma(\frac{\partial^2 \Pi}{\partial \sigma_{atm}^2})$  is significant for certain variations of autocalls.

## 7. Autocalls in a local volatility model: a source of negative carries

Let's have a look at the **ExtraVannas of our CancellablePut** in a local volatility model, split by market data:

| market data $\alpha_k$    | Sign of $\frac{\partial Survival_{rate}(t,T)}{\partial S_t} \frac{\partial PutD\&IPrice}{\partial \alpha_k}$ | Covariance( $Spot, \alpha_k$ ) |                                       | Sign of ExtraVanna P&L  |
|---------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------|---------------------------------------|-------------------------|
|                           |                                                                                                              | Average historical realisation | in local volatility model             |                         |
| <b>Forward</b>            |                                                                                                              |                                |                                       |                         |
| dividend                  | +                                                                                                            | positive                       | dividend split between cash and yield | adequate specific model |
| repo                      | +                                                                                                            | positive                       | zero                                  | -                       |
| rate                      |                                                                                                              | positive                       | spot/rate hybrid model                | adequate specific model |
| <b>Volatility surface</b> |                                                                                                              |                                |                                       |                         |
| implied volatility atm    | +                                                                                                            | around 0 (recently)            | negative                              | -                       |
| implied skew              |                                                                                                              | negligible                     | zero                                  | 0                       |
| implied convexity         | -                                                                                                            | negative                       | zero                                  | -                       |
| <b>Multi-Underlyings</b>  |                                                                                                              |                                |                                       |                         |
| equity correlation        | -                                                                                                            | negative                       | zero                                  | -                       |
| quanto drift              | -                                                                                                            | negative                       | zero                                  | -                       |

As one sees, ExtraVanna P&L are negative. Of course, spot/volatility Vanna remains the most obvious loss but, for a mono-underlying autocall, spot/repo or spot/convexity vanna P&Ls arise now as serious troubles for exotic books.

Are we able to evaluate these expected losses ? We expose, in the following chapter, a simple methodology to price them and replicate them.

## 8. A Trading recipe to price and hedge the model-dependent ExtraVanna P&L of autocalls

The formula which gives the value of the **ExtraVanna P&Ls can be seen as a pay-off**:

$$ExtraVannaP\&L_{0 \rightarrow T} = \int_0^T I|_{no\ recall\ before\ t} \sum \alpha_k \frac{\partial Survival_{rate}(t,T)}{\partial S_t} \frac{\partial PutD\&IPrice}{\partial \alpha_k} \cdot [RealisedCov(dS_t, d\alpha_k) - ModelCov(dS_t, d\alpha_k)|_t] \quad (9.)$$

The formula above is a pay-off which delivers value every business day; this analytical value depends on:

- A Boolean which is worth 1 if no recall occurred before the concerned day
- $Survival_{rate}(t, T)$  and  $PutD\&IPrice$  which are closed formula
- $RealisedCov(dS_t, d\alpha_k)$  can be set according to historical (statistical) values and risk appetite.

One can even consider to set up a function depending on  $S_t$ , if the realised covariance is influenced by the level of spot (which could be interpreted as a skew of covariance).

- $ModelCov(dS_t, d\alpha_k)$  which is null, except for spot/vol covariance which requires a calculation (see *Annex 1*)

In our suggested setup, this pay-off (9.) is booked as an add-on to an autocall priced in a local volatility model. This additional pay-off is priced in a local volatility model and will be replicated as the rest of the portfolio. . Its initial price is a component of the total product initial price.

For any  $S_t$  and any business day  $t$ , this add-on delivers on a daily basis to the Trading book an extra positive P&L, which is designed to compensate exactly ExtraVanna losses calculated earlier.

The add-on (9.) is the sum of these daily flows, priced and replicated. Our setup is a dynamic hedging solution for Vanna negative carries due to the model-dependency of autocalls.

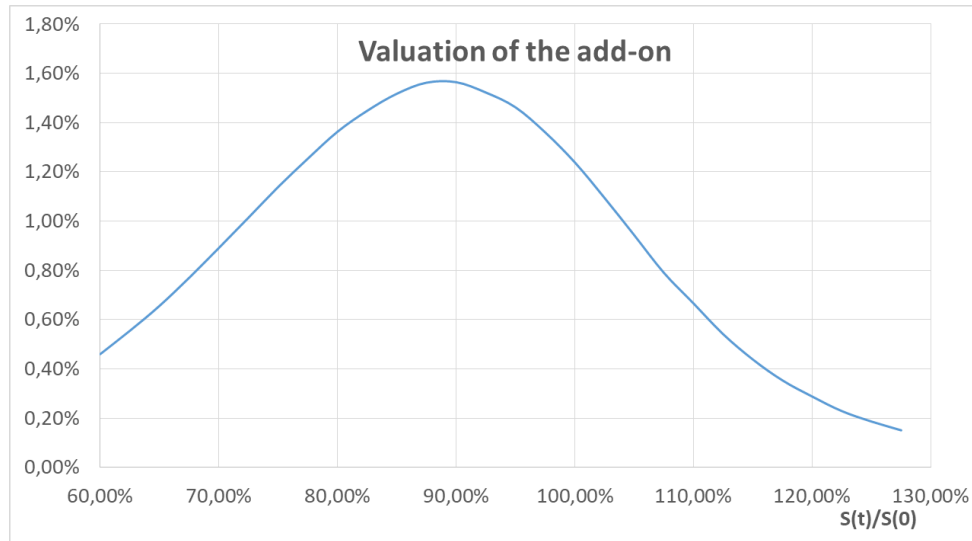
### 9. A standard 10 years autocall on SX5E: behaviors of *ExtraVannaP* & $L_{0 \rightarrow T}$

Let's look at a standard autocall on Eurostoxx50E:

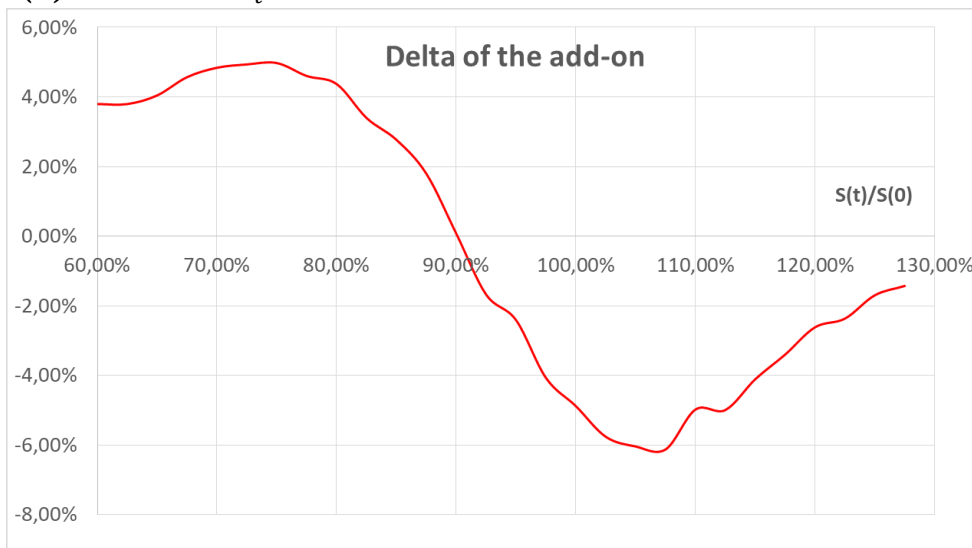
- Striking at date  $t_0$ ,
- European Down&In put at 50% maturity in T,  $T = t_0 + 10\text{years}$
- Recalled if at an anniversary date  $t_i$ ,  $SX5E(t_i) > SX5E(t_0)$ ,

The pricing is computed at date  $t = t_0 + 1\text{day}$

**Valuation in t of the add-on (9.) *ExtraVannaP* &  $L_{t \rightarrow T}$  in function of  $S_t$**



**Delta in t of the add-on (9.) in function of  $S_t$**



Our add-on has to be compared with the difference of price found between stochastic volatility model and local volatility model applied to the autocall specified above.

The graphs exhibit behaviors which are very similar to those exhibited by an efficient stochastic volatility model.

- At inception, when the spot is equal to 100%, the price is around 1,20% and the delta to be sold is worth 4,5%
- The maximum add-on valuation arises when the spot is around 85%
- The add-on pricing converges to zero for high and low spots



## 10. Conclusion

The cancellation feature on an autocall explains why the model-dependency manifests itself through Vanna terms: when spot moves, it needs dynamic rehedging with vanilla options.

One found closed formulas to approximate the Vanna derivatives and, by consequence, an analytical expression of the 'ExtraVannas P&L'. This pay-off is equivalent to the future expected negative carries due to the model-dependency of autocalls.

This proxy valuation gives an efficient, dynamic and simple hedging solution which addresses toxic covariances between spot and volatility (skew and convexity), dividends, repo, quanto drift...

Of course, this empirical method relies on approximations but it shows many advantages:

- It is built on the understanding of P&L leakages: it delivers a daily cash-flow compensating the local Vanna loss.
- Since it is not model based, it does not involve any complex recalibration of tradable instruments, which, in practice, could lead to inconsistencies.
- One keeps the entire control of what is computed.
- Inputs are easy to evaluate: past covariances can be estimated using historical data.
- It fits the empirical knowledge about the instability and dynamic rehedging of an autocall, which could be summarized as follows: the spot moves, the probability of being recalled drops and the trader has to sell an additional quantity of vanilla Down&In puts.
- It is very easy and cheap to implement in pre-existing setups which provide a flexible pay-off scripting language and a Monte Carlo based local volatility pricer.
- Generalizing to multi-underlyings worst-of type of products requires minimal adaptations.

We would like to mention here potential areas of improvement to the setup described in this paper and remaining risks.

- The add-on replication itself might be less than perfect: the studied back-tests show a worst case scenario where 20% of the add-on value fails at being replicated.
- Traders are ultimately market-makers of "realized variances" the same way a vanilla trader would be a market-maker of realized volatility while showing a price on an underlying over which no options are traded.
- One needs to take into account the dynamic hedging of the incremental coupons of the autocall. There also exists a closed formula which allows to incorporate this aspect.

### About the author

Gregory Salon works as a senior consultant for Quantik Trading Consulting; he provides quantitative solutions dedicated to Exotic Trading activities of investment banks.

He warmly thanks Lorenzo Bergomi for his countless advice and fruitful discussions and Selim Mehrez for his constant support and interest about this subject. He shows gratitude to proofreaders: Lorenzo Bergomi and, of course, his partners in Quantik: F Bersani, S Kermiche, N Mahrour.

## ***Annex 1: Local volatility model, an embedded covariance between spot and volatility***

Let's define  $\hat{\sigma}_{K,T}$  as the implied volatility (strike K, maturity T),  $\mathcal{S}_T$  as the implied skew and  $F_T$  as the forward of maturity T. We express the skew around the money forward as:

$$\mathcal{S}_T = \left. \frac{d\hat{\sigma}_{K,T}}{d \ln K} \right|_{F_T}$$

We use the following result (Bergomi Lorenzo, *Stochastic Volatility Modeling*, Chapman and Hall, 2016, p. 51):

$$\frac{d\hat{\sigma}_{F_T,T}}{d \ln S_0} = \mathcal{S}_T + \frac{1}{T} \int_0^T \frac{\bar{\sigma}^2(u)}{\hat{\sigma}_u \hat{\sigma}_T} \cdot \mathcal{S}_u du$$

With  $\bar{\sigma}^2(u)$  allowing the calibration of the term structure of forward volatility through  $\hat{\sigma}_{F_T,T} = \sqrt{\frac{1}{T} \int_0^T \bar{\sigma}^2(u) du}$

As a consequence, it comes:

$$LocalVolModelCovariance(d \ln S_0, d\hat{\sigma}_{F_T,T}) = \hat{\sigma}_{F_T,T} \cdot \frac{d\hat{\sigma}_{F_T,T}}{d \ln S_0} = \hat{\sigma}_{F_T,T} \cdot \left[ \mathcal{S}_T + \frac{1}{T} \int_0^T \frac{\bar{\sigma}^2(u)}{\hat{\sigma}_u \hat{\sigma}_T} \cdot \mathcal{S}_u du \right]$$